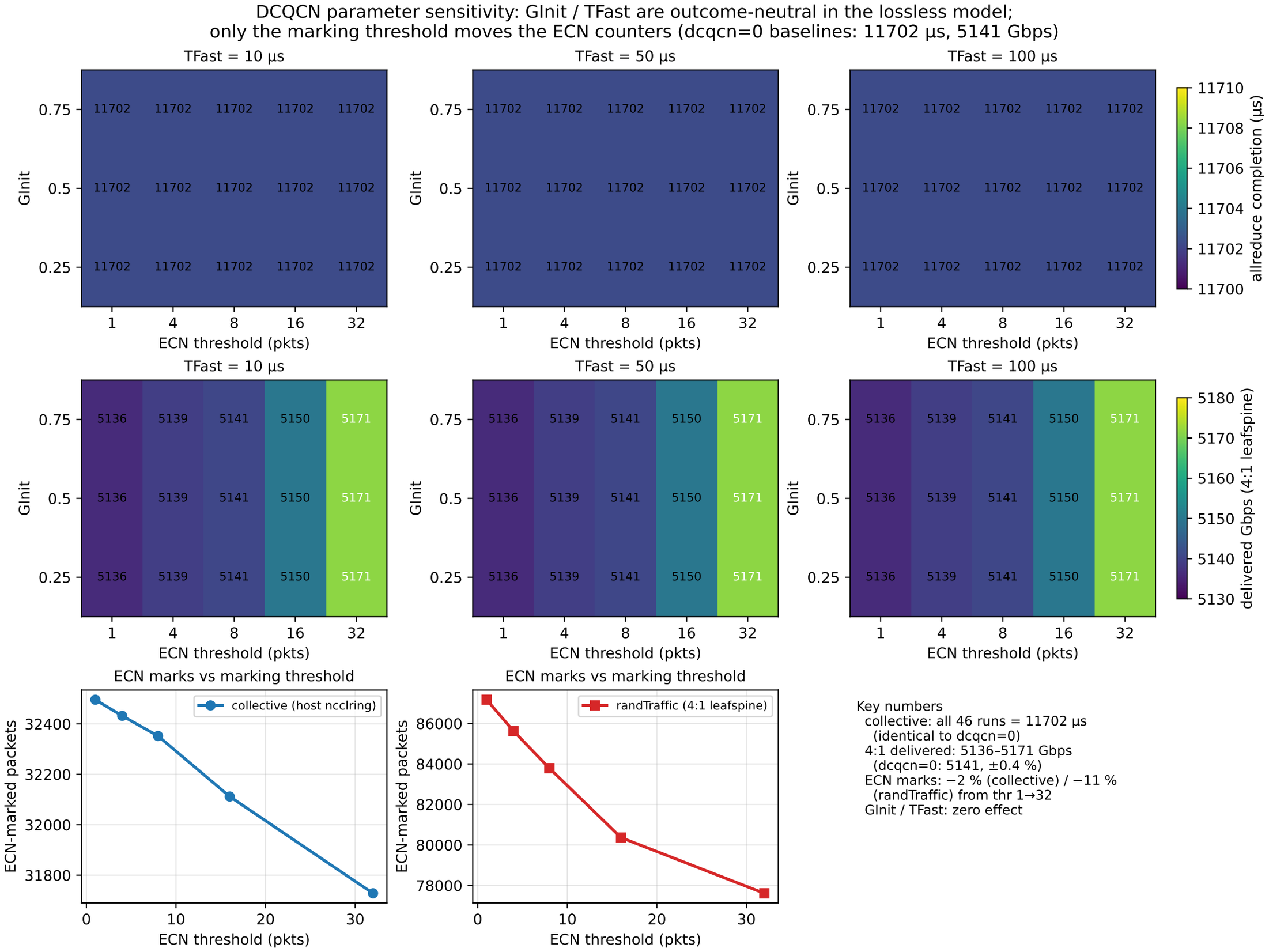
DCQCN parameter sensitivity: GInit / TFast are outcome-neutral in the lossless model;
only the marking threshold moves the ECN counters (dcqcn=0 baselines: 11702 µs, 5141 Gbps)
TFast = 10 µs
TFast = 50 µs
TFast = 100 µs
GInit
ECN threshold (pkts)
allreduce completion (µs)
delivered Gbps (4:1 leafspine)
ECN marks vs marking threshold
ECN-marked packets
collective (host ncclring)
randTraffic (4:1 leafspine)
Key numbers
collective: all 46 runs = 11702 µs
(identical to dcqcn=0)
4:1 delivered: 5136–5171 Gbps
(dcqcn=0: 5141, ±0.4 %)
ECN marks: −2 % (collective) / −11 %
(randTraffic) from thr 1→32
GInit / TFast: zero effect